

Fractal Cascade: From Proton to Solar Photosphere

Universal Kochenov Framework (UKF)

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August 2026

1. Cascade

$$\boxed{\lambda_p \xrightarrow{\times \Lambda} R_{\text{core}} \equiv a_{\text{Io}} \xrightarrow{\times \alpha^{-1}} a_{\text{Mercury}} \xrightarrow{\times S} \bar{\lambda}_e}$$

$$\Lambda = \frac{2 \times 144^3 \times 31^2}{720 \times 2.5} \times 10^{17} = 3.188367 \times 10^{23}$$

$$S = \frac{720 \times 2.5}{2 \times 144^3 \times 31^2} \times 10^{14} = 31\,364 \text{ km}$$

S — Space Step, the lattice spacing

$$\Lambda_e = \frac{a_{\text{Mercury}}}{\bar{\lambda}_e} = 1.49962 \times 10^{23}$$

2. Operator Register

$$\mathcal{M} = \{31, 144, 720, \phi, \sqrt{5}, \alpha^{-1}, \chi\} \quad (F_{12} = 144 \in \text{Fibonacci})$$

$$\chi_{\text{base}} = \frac{720}{2 \cdot 144} = 2.5, \quad \chi \equiv \frac{m_n - m_p}{m_e} \approx 2.531$$

$$\frac{M_5}{M_7 - M_5} = \frac{31}{96}, \quad \frac{M_5^2}{M_7 - M_5} = \frac{961}{96} = 10.010416$$

$$\frac{M_5^2}{F_{12}} = \frac{31^2}{144} = \frac{961}{144} = 6.673611 \quad (\delta = -0.01\%)$$

$$\alpha^{-1} = 137.036, \quad M_5 = 31, \quad M_7 = 127$$

3. Proton $\rightarrow$ Io

$$\lambda_p = 1.321409855 \times 10^{-15} \text{ m}$$

$$\Lambda_{\text{UKF}} = 3.188367 \times 10^{23}$$

$$R_{\text{core}} \equiv a_{\text{Io}} = \lambda_p \times \Lambda_{\text{UKF}} = 4.21314 \times 10^8 \text{ m} = 421\,314 \text{ km}$$

Verification:

JPL/NASA mean semi-major axis of Io: 421 800 km

$$\delta = +0.115\%$$

Independent Io estimates obtained through the proton and electron branches.

4. Electron $\rightarrow$ Mercury

$$\bar{\lambda}_e = 3.8615926744 \times 10^{-13} \text{ m}$$

$$a_{\text{Mercury}} = 0.38709893 \text{ AU} = 5.7909175678 \times 10^{10} \text{ m}$$

$$\Lambda_e = \frac{a_{\text{Mercury}}}{\bar{\lambda}_e} = \frac{5.7909175678 \cdot 10^{10}}{3.8615926744 \cdot 10^{-13}} = 1.49962 \cdot 10^{23}$$

$$\text{Verification: } \frac{\Lambda_e \cdot 10^{-12}}{1 \text{ AU}} = \frac{1.49962 \cdot 10^{11}}{1.495978707 \cdot 10^{11}} = 1.00243 \quad (\delta = +0.243\%)$$

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$$\Delta_{\text{hadron}} = a_{\text{Mercury}} - \bar{\lambda}_e^{\text{norm}} = 0.387099 - 0.386159 = 0.000939 \quad (\equiv m_n \cdot 10^{-6} \text{ MeV})$$

5. Mercury $\rightarrow$ Io $\rightarrow$ Europa $\rightarrow$ Ganymede $\rightarrow$ Callisto

$$\begin{pmatrix} a_{\text{Io}} \\ a_{\text{Europa}} \\ a_{\text{Ganymede}} \\ a_{\text{Callisto}} \end{pmatrix} = \frac{a_{\text{Mercury}}}{\alpha^{-1}} \cdot \begin{pmatrix} 1 \\ \sqrt{\chi} \\ \chi \\ \phi^3 + \frac{31}{137} \end{pmatrix}$$

$$\frac{a_{\text{Mercury}}}{\alpha^{-1}} = \frac{57\,909\,176}{137.036} \approx 422\,583 \text{ km}$$

Table 1: Macroscopic Projection

Satellite	UKF (km)	JPL/NASA (km)	Deviation
Io	422 583	421 800	+0.19%
Europa	672 292	671 100	+0.18%
Ganymede	1 069 557	1 070 400	−0.08%
Callisto	1 885 711	1 882 700	+0.16%

6. Io – Pan – Daphnis – Atlas – Titan

$$\begin{pmatrix} a_{\text{Pan}} \\ a_{\text{Daphnis}} \\ a_{\text{Atlas}} \end{pmatrix} \equiv a_{\text{inner Saturn}} = a_{\text{Io}} \cdot \frac{M_5}{M_7 - M_5} = a_{\text{Io}} \cdot \frac{31}{96} \approx 136\,458 \text{ km}$$

Table 2: Saturnian Cascade

Satellite	UKF (km)	JPL/NASA (km)	Deviation
Pan	136 458	133 584	+2.15%
Daphnis	136 458	136 505	−0.03%
Atlas	136 458	137 670	−0.88%
Mean	136 458	135 920	+0.40%

$$a_{\text{Io}} = a_{\text{Titan}} \times \frac{\sqrt{3}}{5} = 1\,221\,870 \text{ km} \times 0.346410 = 423\,268.18 \text{ km} \quad (\delta = +0.37\%)$$

Uranus Reverse Projection

$$a_{\text{ring}} = a_{\text{Titania}} \times \frac{31^2}{2^{13} - 1} = 435\,910 \times \frac{961}{8191} = 51\,143 \text{ km}$$

Observed ε -ring core radius: 51 149 km. Deviation: −0.012%.

The same register ratio that appears in the inner cascades also recovers the principal ring scale of Uranus.

Venus: Inner-System Inversion

$$a_{\text{Venus}} = 1 \text{ AU} \times \frac{\varphi}{\sqrt{5}} = 0.723607 \text{ AU}$$

Observed semi-major axis of Venus (NASA): 0.723332 AU. Deviation: +0.038%.

The same register ratio that appears in the inner cascades also recovers the orbit of Venus.

Golden Bridge (Photospheric Projection)

$$B_\phi = \varphi + \frac{31}{1000} = 1.649034$$

$$R_{\odot}^{\text{pred}} = R_{\text{core}} \times B_\phi = 421\,560 \times 1.649034 = 695\,167 \text{ km}$$

Empirical bridge coefficient 31/1000

Modern helioseismic and transit measurements determine the actual physical radius of the solar photosphere as:

$$R_{\odot} \approx 695\,700 \text{ km}$$

The relative deviation is:

$$\delta = \frac{695\,167 - 695\,700}{695\,700} \approx -0.076\%$$

Deviation: -0.076% . The operator links the proton-derived core scale to the visible photosphere.

System	Operator	UKF Prediction	Deviation
Uranus ϵ -ring	$a_{\text{Titania}} \times \frac{31^2}{2^{13} - 1}$	51 143 km	-0.012%
Venus orbit	$1 \text{ AU} \times \frac{\phi}{\sqrt{5}}$	0.723607 AU	$+0.038\%$
Solar photosphere	$R_{\text{core}} \times \left(\phi + \frac{31}{1000} \right)$	695 167 km	-0.076%

Electron–Mercury Correspondence and the Origin of the Astronomical Unit

Within the UKF address hierarchy the Sun is assigned the role of the central proton-scale object and Mercury the role of the corresponding electron-scale object. The numerical content of this assignment is expressed by the ratio of Mercury’s semi-major axis to the reduced Compton wavelength of the electron:

$$\begin{aligned}\bar{\lambda}_e &= 3.8615926744 \times 10^{-13} \text{ m}, \\ a_{\text{Mercury}} &= 5.7909175678 \times 10^{10} \text{ m}.\end{aligned}$$

Their quotient defines the pure dimensionless address

$$\Lambda_e = \frac{a_{\text{Mercury}}}{\bar{\lambda}_e} = 1.49962 \times 10^{23}. \quad (1)$$

Rescaling by twelve orders of magnitude recovers the astronomical unit:

$$\frac{\Lambda_e \times 10^{-12}}{1 \text{ AU}} = 1.00243 \quad (\delta = +0.243 \%). \quad (2)$$

The electron–Mercury correspondence is *not* introduced as a derivation of Mercury’s orbit from the electron Compton wavelength. Rather, the observed Mercury semi-major axis defines a dimensionless address when expressed in reduced electron-Compton units. This address then exhibits a numerical correspondence with the astronomical unit under the framework’s 10^{-12} rescaling:

$$\bar{\lambda}_e \longrightarrow \Lambda_e = \frac{a_{\text{Mercury}}}{\bar{\lambda}_e} \longrightarrow \Lambda_e \times 10^{-12} \approx 1 \text{ AU}. \quad (3)$$

Once the Mercury address is fixed, the same register operators (α^{-1} , χ , Mersenne ratios) generate the Galilean satellite hierarchy. Two independent geometric routes yield the Io address:

- **Proton branch:**

$$a_{\text{Io}} = \lambda_p \times \Lambda_{\text{UKF}} = 421\,314 \text{ km} \quad (\delta = -0.115 \%).$$

- **Electron branch:**

$$a_{\text{Io}} \approx \frac{a_{\text{Mercury}}}{\alpha^{-1}} \approx 422\,583 \text{ km} \quad (\delta = +0.19 \%).$$

The two routes, starting from the proton Compton wavelength and from the electron Compton wavelength respectively, converge on the same physical orbit to within $\sim 0.3\%$.

The construction therefore supplies an explicit numerical bridge between the electron Compton scale, the Mercury address, the astronomical unit and the Io address — all remaining geometric and employing only the dimensionless address together with the integer register set of the framework.

Related Works

This paper is part of the Universal Kochenov Framework (UKF) series:

- **Paper 1:** Hydrogen Level Addresses in the UKF Kinematic Framework: A Geometric Rewriting of the Bohr Scale and the $1/n^2$ Spectral Skeleton
[10.5281/zenodo.21940531](https://zenodo.org/record/21940531)
- **Paper 2:** Universal Kochenov Framework v4.8.5: A Kinematic-Geometric Description of Orbital Dynamics, Relativistic Perihelion Precession, and Solar Light Deflection
[10.5281/zenodo.21940484](https://zenodo.org/record/21940484)
- **Paper 3:** A Mass-Free Kinematic Topology for Orbital Mechanics
[10.5281/zenodo.21940519](https://zenodo.org/record/21940519)

All papers are available as preprints on Zenodo under the UKF series.